Module – 5
Traffic Management Systems For Safety

1. Road Safety Audit (RSA):

A Road Safety Audit is a formal, independent, multidisciplinary examination of an existing or proposed road, intersection, or road network to identify potential safety issues for all types of road users (vehicles, pedestrians, cyclists) — before these issues result in crashes. It does not wait for accidents to occur; instead, it proactively assesses design and operational safety.

1.1 Purpose / Why RSA is Important:

- *To find safety problems early:* RSAs help identify things that can cause accidents, such as poor visibility, bad road design, confusing signs, or lack of pedestrian safety.
- *To suggest improvements:* The audit team gives recommendations like adding signs, improving curves, widening footpaths, adding speed control, etc., to prevent accidents before they happen.
- *To make roads safer overall:* RSAs help reduce the number and severity of crashes, even on new roads where accidents have not yet occurred.
- *To include safety from the beginning:* Safety becomes part of every stage—planning, design, construction, and after opening—so safety is built-in from day one, not fixed later.

1.2 Who Performs It & What Makes RSA Different:

A typical RSA team has 3–5 independent members with different areas of expertise:

1. Team Leader (Road Safety Engineer)

- Leads and coordinates the audit.
- Ensures the audit is properly carried out and the report is accurate.
- Must have good experience in road design and safety.

2. Traffic / Transportation Engineer

- Studies traffic flow, intersections, and signal control.
- Understands how drivers behave in real traffic conditions.

3. Highway / Road Design Engineer

- Examines road alignment, curves, signs, drainage, cross-sections, etc.
- Checks whether design features meet safety standards.

4. Human Factors Specialist

- Studies driver behaviour, perception, reaction time, fatigue, etc.

- Helps understand how human limitations affect road safety.

5. Local Police / Maintenance Personnel

- Provide real-world insights into issues like speeding, signage problems, visibility, or maintenance issues.

Qualifications of Road Safety Auditors

Auditors should have:

1. Education

- Degree in civil engineering, traffic engineering, transportation planning, or related fields.

2. Experience

- Practical experience in road design, traffic operations, crash analysis, or safety projects.

3. Training / Certification

- Many agencies require completion of formal RSA training programs.

4. Knowledge of Standards

- Should be familiar with national/international road design standards, safety guidelines, signage rules, etc.

Responsibilities of RSA Auditors (Clean & Clear)

1. Preparation

- Study project documents, drawings, previous audits, and traffic data.
- Understand road environment, land use, and expected road users.

2. Site Visit

- Observe actual road conditions, lighting, traffic behaviour, markings, and visibility.
- Identify risks that may not appear on drawings.

3. Safety Issue Identification

- Detect design or operational elements that may cause accidents.
- Consider all road users—drivers, pedestrians, cyclists, elderly, children, etc.
- Assess every issue objectively.

4. Reporting

- Prepare a clear audit report listing safety problems and giving practical recommendations.
- Ensure suggestions are realistic and prioritised based on risk.

5. Communication

- Discuss findings with project owners or designers.

- Maintain professionalism and neutrality.

2. Tools for Safety Management Systems:

A Safety Management System in road transport refers to a structured, data-driven framework used to identify, analyze, and reduce safety risks on road networks. It uses several specialized tools to support continuous monitoring and improvement. The major tools used under SMS include:

a) Crash Data Analysis Systems

These systems collect and analyze crash data from police reports, hospitals, and road agencies. By studying crash patterns, high-risk locations (black spots), vehicle types involved, and contributing factors, planners can identify locations needing immediate intervention. Crash analysis supports evidence-based decision-making and enables trend monitoring for long-term safety planning.

(b) Road Safety Impact Assessments (RSIAs)

RSIAs are carried out during the planning stage of major road projects. They evaluate the possible safety impacts of new developments or changes in land use and traffic patterns. RSIAs help planners foresee future risks and integrate corrective measures early in the design. This prevents creation of new hazards and ensures sustainable, safe road infrastructure from the beginning.

(c) Road Safety Inspections

These are regular, on-site evaluations of existing roads to detect hazards such as poor pavement conditions, missing signage, inadequate lighting, potholes, or faded markings. Unlike audits, inspections focus on operational roads and support **continuous safety monitoring**. They enable timely maintenance and prevent accidents caused by infrastructure failures.

(d) Safety Performance Indicators (SPIs)

SPIs are measurable indicators used to track safety performance over time. Examples include crash rate per kilometre, fatality index, pedestrian exposure, and vehicle speed distribution. SPIs help authorities set realistic safety goals and evaluate whether interventions under the SMS are effective. They promote data-driven evaluation and facilitate continuous improvement.

3. Road Safety Audit (RSA) Process

RSA process is explained in below steps:

1. Selection of Audit Team

- An independent, qualified, multidisciplinary team is appointed.
- Usually includes road engineers, traffic safety experts, enforcement officers, and pedestrian/cyclist specialists.
- Independence ensures unbiased safety evaluation.

2. Project Briefing & Collection of Documents

- Audit team receives all relevant design documents:
    - Drawings, plans, layouts
    - Traffic studies, speed data
    - Land-use information
- Purpose is to understand the project fully before inspection.

3. Initial Review of Design

- Team checks design for potential safety issues like:
    - Sharp curves
    - Insufficient sight distance
    - Missing pedestrian provisions
    - Improper signage or lane width
- Helps identify early-stage risks.

4. Site Inspection / Field Visit

- Team visits the road site (existing or proposed alignment).
- Inspections are done during day and night to assess conditions under different visibility.
- Team checks:
    - Road geometry
    - Pedestrian movement
    - Surrounding land-use
    - Lighting
    - Traffic behaviour (if road is already operating)
- Photos and measurements are taken for evidence.

5. Identification of Safety Issues

- Team lists all potential hazards observed from documents and site visit.

- Issues may relate to:
    - Geometry
    - Visibility
    - Intersections
    - Pedestrian/cyclist safety
    - Speed and traffic flow
- Each issue is analysed for how it can contribute to crashes.

6. Development of Recommendations

- Team suggests practical and low-cost improvements such as:
    - Adding signage, markings, guardrails
    - Improving lighting and sight distance
    - Redesigning intersections
    - Providing footpaths or crossings
- Recommendations focus on preventing crashes before the road opens or before accidents occur.

7. Preparation of RSA Report

- A formal report is prepared that includes:
    - Description of the project
    - All identified safety issues
    - Photographs, sketches
    - Recommended corrective measures
- Report is submitted to the client/road authority.

8. Client Response / Decision Making

- Project owner reviews the recommendations.
- Decides which ones will be implemented.
- A Response Report is prepared explaining accepted, modified, or rejected recommendations.

9. Implementation of Safety Improvements

- Approved recommendations are implemented during:
    - Planning stage
    - Design stage
    - Construction stage

        - Operational stage

10. Follow-up Audit / Post-Construction Review

- Conducted to ensure:
    - Recommendations were implemented correctly
    - No new safety issues have emerged
- Helps verify the effectiveness of safety improvements.

4. Road Safety Improvement Strategies:

Road safety improvement strategies include all measures taken to reduce crashes, minimise injuries, and create safer roads for all users. These strategies focus on engineering improvements, enforcement, education, emergency care, and management practices.

1. Engineering Measures

These involve improving the physical road environment:

- Proper road design, lane width, shoulders, curves, and sight distance.
- Junction improvements like signals, roundabouts, channelization.
- Speed-management through humps, rumble strips, and appropriate speed limits.
- Better road surface quality, skid-resistance, and drainage.
- Safe pedestrian and cyclist facilities such as footpaths, crossings, medians, lighting.

2. Enforcement Measures

- Strict enforcement of speed limits, drunk driving, helmet/seat-belt rules, and mobile-phone restrictions.
- Strong penalty system to discourage unsafe behaviour.
- Vehicle fitness checks and control of overloading.

3. Education & Awareness

- Road safety education in schools, colleges, and communities.
- Media campaigns promoting safe speeds, avoiding alcohol, and safe pedestrian behaviour.

4. Emergency Response

- Quick ambulance access and improved trauma care reduce fatality rates.
- Training first responders and clearing crash sites quickly.

5. Road Safety Management

- Road Safety Audits, black-spot identification, and safety inspections.
- Continuous monitoring using crash data and performance indicators.

6. Technology-Based Measures

- Use of CCTV, speed cameras, red-light cameras, and ITS systems.
- Vehicle technologies like ABS, airbags, and lane-assist.

5. ITS and Safety:

Intelligent Transportation Systems (ITS) refer to the use of advanced technologies—such as sensors, cameras, communication systems, and computer-based control—to improve the efficiency and safety of road transportation. ITS helps reduce crashes, improve traffic flow, support enforcement, and provide timely information to road users. It makes roads safer by detecting risks early, warning drivers, and supporting better traffic management.

1. Real-Time Traffic Monitoring

ITS uses CCTV cameras, traffic sensors, GPS data, and automatic number plate recognition to monitor road conditions continuously.

This helps in:

- Detecting accidents instantly
- Identifying traffic jams
- Sending alerts to control centres. Quick detection reduces secondary crashes and allows faster emergency response.

2. Speed Management

ITS systems such as:

- Speed cameras
- Radar-based speed display boards
- Variable speed limit signs - help control vehicle speeds. Maintaining correct speed reduces severity of crashes and helps protect pedestrians and cyclists.

3. Red-Light Violation Detection

Automatic Red-Light Cameras record vehicles that jump signals. This improves safety at intersections by discouraging risky behaviour, reducing severe side-impact ("T-bone") crashes.

4. Lane Enforcement and Wrong-Way Detection

Systems detect:

- Lane violations

- Wrong-way driving
- Heavy vehicles entering restricted lanes

  Warnings are issued immediately to drivers and traffic police. This prevents head-on collisions and improves discipline on highways.

5. Pedestrian and Cyclist Safety

ITS supports vulnerable road users by:

- Smart pedestrian crossings (signals activate only when people are present)
- Automatic flashing lights when people enter zebra crossings
- Thermal or motion sensors near school
- These reduce pedestrian crashes significantly.

6. Traffic Signal Coordination

ITS synchronizes traffic lights using real-time traffic flow data. Benefits:

- Smoother traffic
- Fewer sudden stops
- Reduction in rear-end collisions

  This is especially effective in busy urban corridors.

7. Incident and Emergency Management

ITS provides:

- Automatic crash detection
- Quick alert to ambulances and police
- GPS-based vehicle tracking
- Display of warnings on digital signboards.

  Fast response reduces fatalities and prevents secondary crashes.

8. Driver Assistance Technologies

ITS includes in-vehicle systems like:

- Anti-lock braking system (ABS)
- Electronic Stability Control (ESC)
- Lane Departure Warning (LDW)
- Collision Avoidance System
- Blind-spot detection

  These help drivers make safer decisions and avoid human errors.